SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Name: Vaishnavi More

Roll Number: B - 72

Batch: SB4

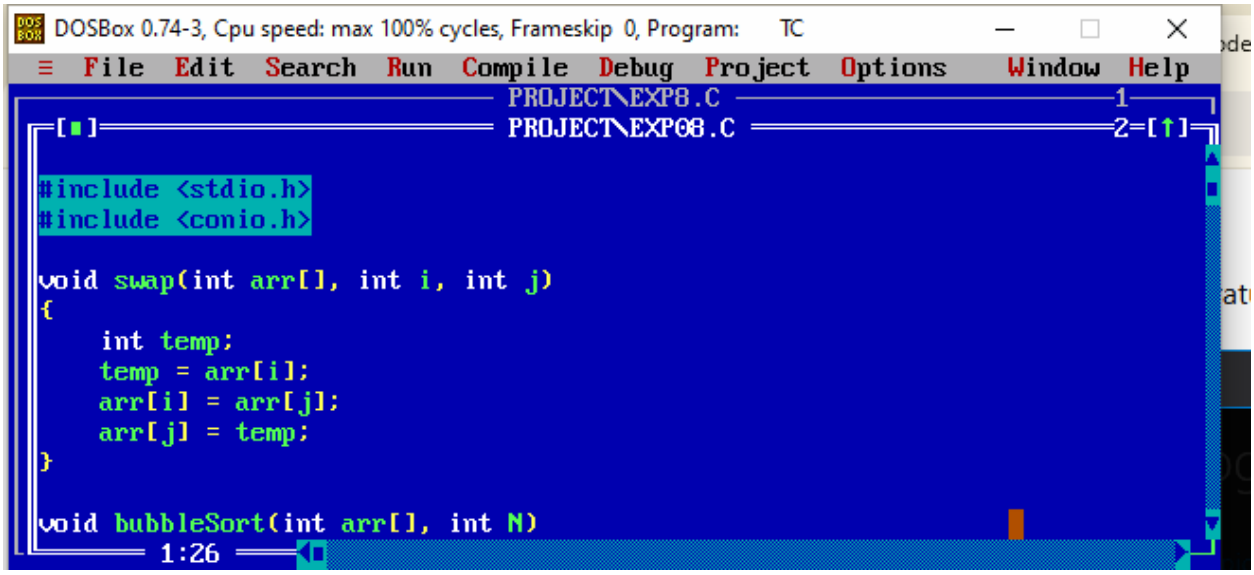
Course Name and Code: Data Structures Lab (ITL302)

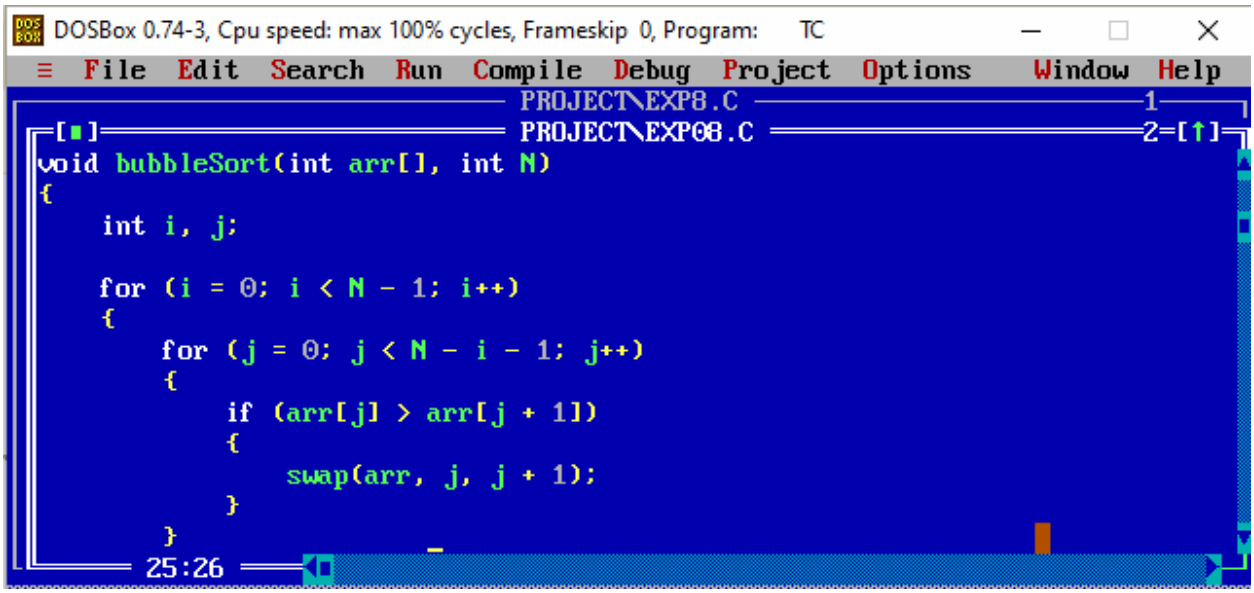
Semester: III (SYIT)

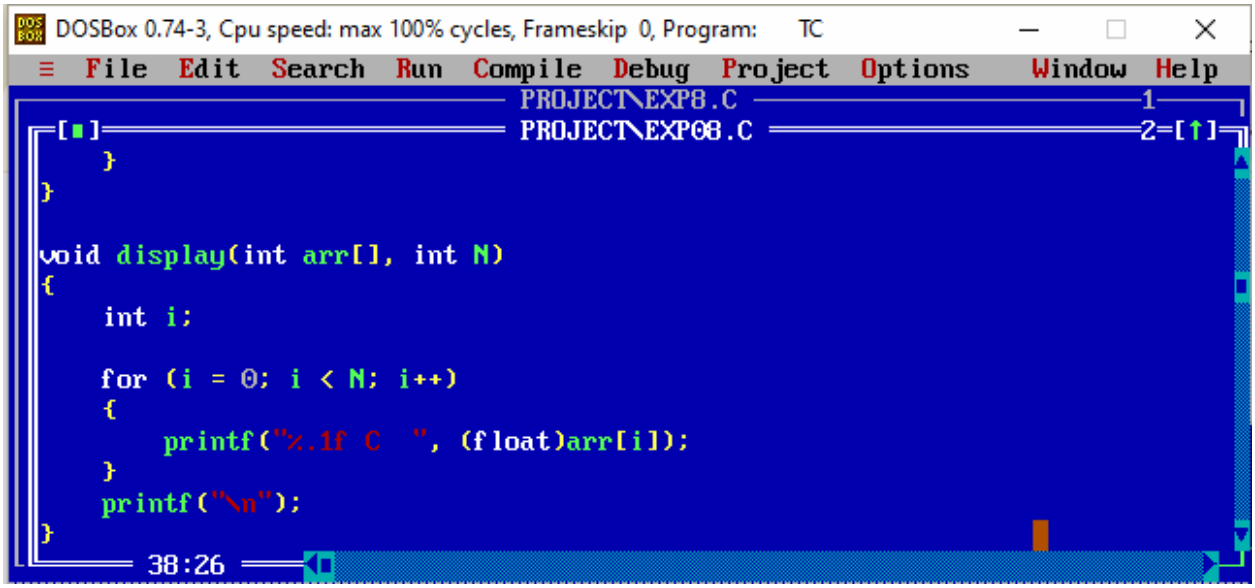
Academic Year: 2026-27 (Odd Semester)

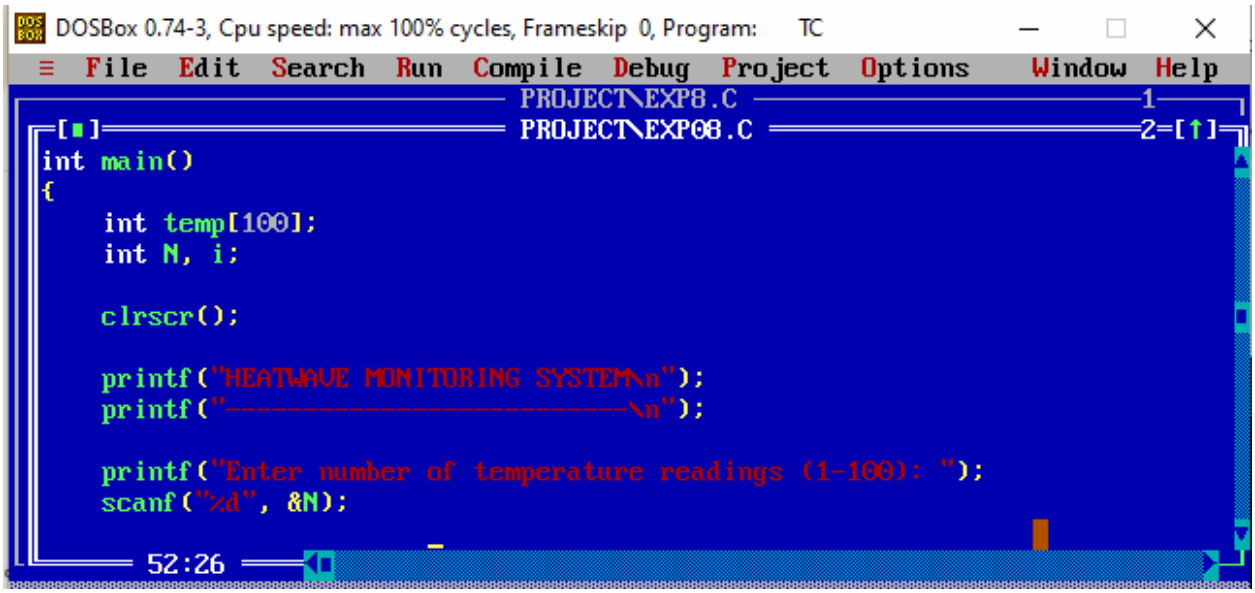
Experiment No. 08

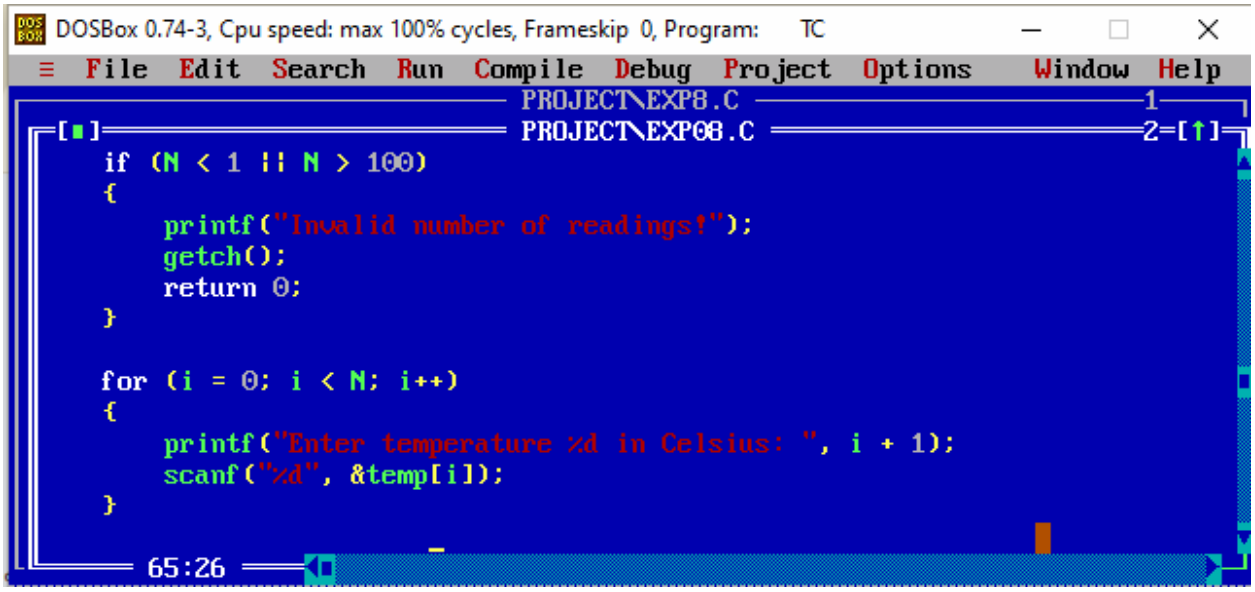
Code:

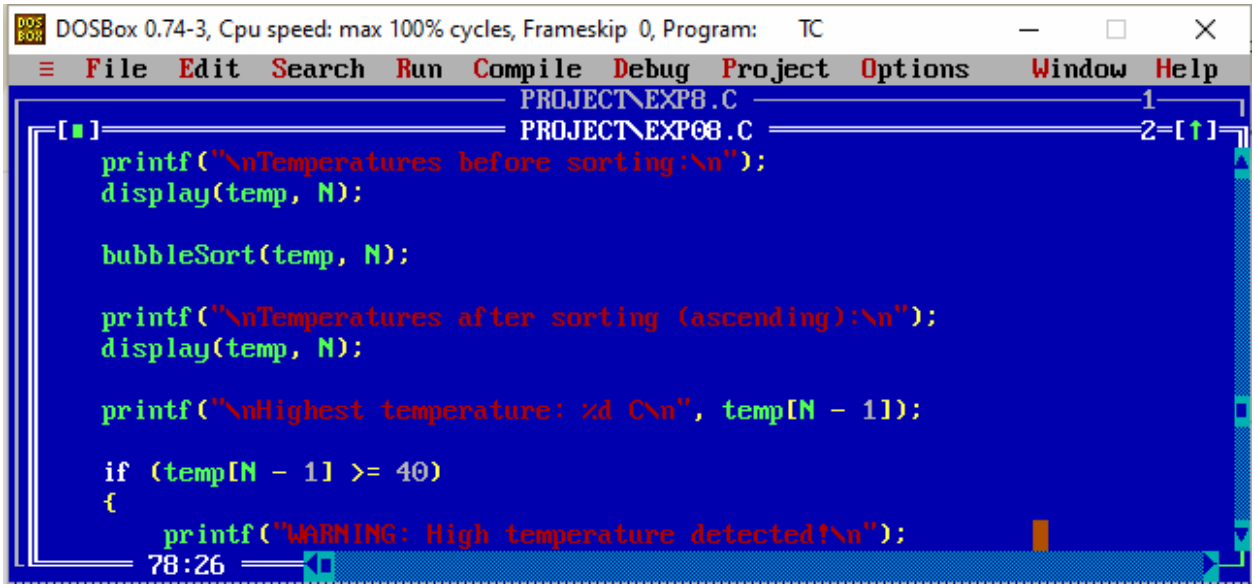


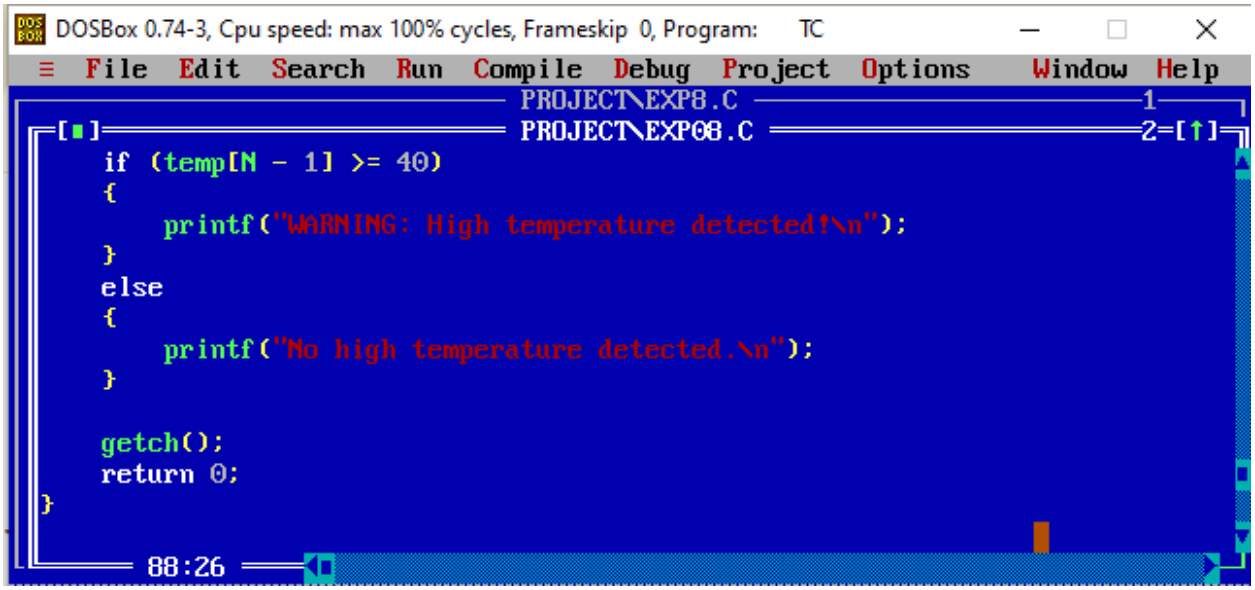

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\EXP8.C
PROJECT\EXP08.C
void bubbleSort(int arr[], int N)
{
int i, j;
for (i = 0; i < N - 1; i++)
{
for (j = 0; j < N - i - 1; j++)
{
if (arr[j] > arr[j + 1])
{
swap(arr, j, j + 1);
}
}
25:26


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\EXP8.C
PROJECT\EXP08.C
}
}
void display(int arr[], int N)
{
int i;
for (i = 0; i < N; i++)
{
printf("%.1f C ", (float)arr[i]);
}
printf("\n");
}
38:26

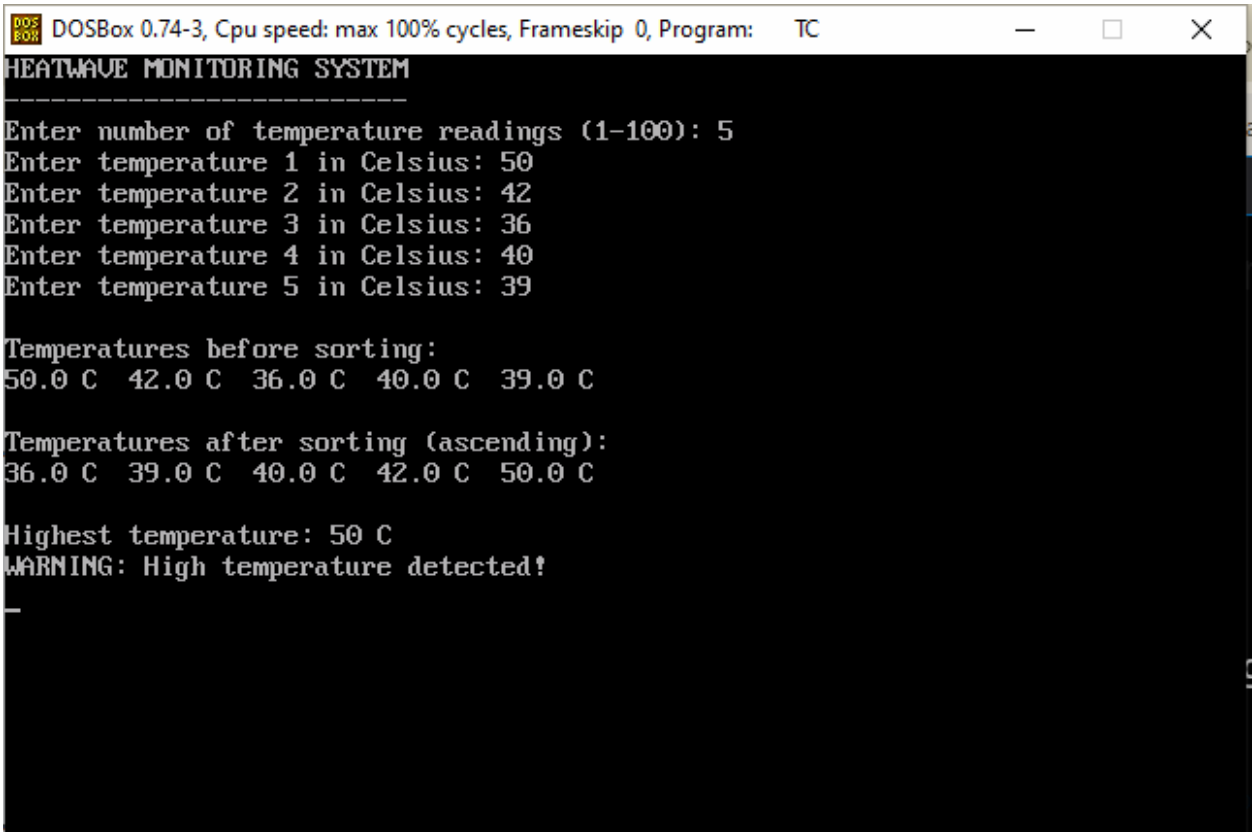

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\EXP8.C 1
PROJECT\EXP08.C 2=[↑]
int main()
{
int temp[100];
int N, i;
clrscr();
printf("HEATWAVE MONITORING SYSTEM\n");
printf("-------------------------\n");
printf("Enter number of temperature readings (1-100): ");
scanf("%d", &N);
52:26


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\EXP8.C 1
PROJECT\EXP08.C 2=[↑]
if (N < 1 || N > 100)
{
printf("Invalid number of readings!");
getch();
return 0;
}
for (i = 0; i < N; i++)
{
printf("Enter temperature %d in Celsius: ", i + 1);
scanf("%d", &temp[i]);
}
65:26


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\EXP8.C
PROJECT\EXP08.C
printf("\nTemperatures before sorting:\n");
display(temp, N);
bubbleSort(temp, N);
printf("\nTemperatures after sorting (ascending):\n");
display(temp, N);
printf("\nHighest temperature: %d C\n", temp[N - 1]);
if (temp[N - 1] >= 40)
{
printf("WARNING: High temperature detected!\n");
78:26


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
File Edit Search Run Compile Debug Project Options Window Help
PROJECT\EXP8.C
PROJECT\EXP08.C
if (temp[N - 1] >= 40)
{
printf("WARNING: High temperature detected!\n");
}
else
{
printf("No high temperature detected.\n");
}
getch();
return 0;
}
88:26

Output:


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
HEATWAVE MONITORING SYSTEM
Enter number of temperature readings (1-100): 5
Enter temperature 1 in Celsius: 50
Enter temperature 2 in Celsius: 42
Enter temperature 3 in Celsius: 36
Enter temperature 4 in Celsius: 40
Enter temperature 5 in Celsius: 39
Temperatures before sorting:
50.0 C 42.0 C 36.0 C 40.0 C 39.0 C
Temperatures after sorting (ascending):
36.0 C 39.0 C 40.0 C 42.0 C 50.0 C
Highest temperature: 50 C
WARNING: High temperature detected!